Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/07/2016
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jyagoobi@wpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	508-831-5759
Organization Name:	Worcester Polytechnic Institute		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	10 W 32nd St, E1 Building, Room 243		
PI Web Page:			
City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2012	**End Date:**	02/28/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2017 per NSSC information (original end date was 9/30/2016)--Ed., 4/23/13		
Key Personnel Changes/Previous PI:	none.		
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX12AR32G		
Performance Goal No.:			
Performance Goal Text:			
	Electrically driven flows can provide significant enhancements to current and future technologies that can benefit from increased heat transfer and mass transport capacities in macro and micro scales. In particular, electrically generated flows (with and without phase change) have great potential for terrestrial and micro-gravity applications. Electrohydrodynamic (EHD) ion-drag pumping and induction pumping have been extensively studied in the past several decades with no significant, successful applications. On the contrary, EHD pumping based on conduction phenomenon has reached a stage where various applications are expected in the near future, since pressure heads on the order of kilo Pascal per electrode pair are easily obtainable. EHD conduction pumps are simple in design, lightweight, non-mechanical, and free of vibration and noise, and they allow for active control of heat transfer and mass transport. EHD conduction pumping requires minimal electric power to operate (at least three orders of magnitude less power		

Task Description:	compared to the resultant heat transport capacity). This proposed research will fundamentally investigate the electrically-driven dielectric liquid film in the presence of phase change in macro- and micro-scales based on conduction phenomenon through a combined experimental and computational investigation. Furthermore, the coupling of electric double layers and heterocharge layers in micro-channels will be explored, and the non-dimensional threshold at which the double layer starts to affect the corresponding flow field will be determined. Note that in micro-scales, the Debye length within dielectric fluids becomes comparable to the conduit dimensions. Additionally, unique two-phase heat transport devices will be developed that will operate based on the conduction phenomenon, with or without capillary effects, in macro- and micro-scales. Heat flux levels with these active micro-scale heat transport devices are expected to significantly exceed the corresponding working fluid critical heat flux level while operating stably.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are a number of potential terrestrial and space applications for this EHD driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices. For further technical and general information, please contact Professor Jamal Yagoobi at jyagoobi@wpi.edu . This device currently is not commercially available.
Task Progress:	The research carried out during this period was based on the combination of the EHD conduction pumping of liquid film toward the heated surface and the extraction of generated vapor bubbles from the heater surface via the dielectrophoretic (DEP) force. Experiments in liquid film flow boiling are conducted in the radial heat transport system configuration. EHD conduction pumping is used to generate liquid film flow from the surroundings for electro-wetting of the heater surface. Simultaneously, a strong non-uniform local electric field above the heater surface is also used to generate the dielectrophoretic force for increased vapor bubble extraction during the nucleate boiling regime. The height of the liquid film and applied potentials for the EHD and DEP electrodes are varied to determine their effect on the entire liquid film flow boiling regime up to critical heat flux. Liquid film flow boiling and pool boiling in the presence of combined EHD conduction pumping and dielectrophoretic force were successfully studied. The results showed remarkable enhancement in heat flux and heat transfer coefficient up to 1217% for a given superheat value when both mechanisms were used. The ratio of electric power input for the combined techniques was only 0.7% compared to the corresponding heater power increase for this level of enhancement. The results also provided new physical insights into the dominant mechanisms of heat transfer enhancement during liquid film flow boiling and pool boiling in the presence and absence of electric fields. For example, they showed that at low heat flux, the EHD conduction-generated liquid film flow allowed for increased forced convection and delayed the onset of nucleate boiling. It was found that increasing the depth of the liquid pool was equivalent to increasing the applied EHD potential, since both resulted in higher pressure within the liquid which resulted in higher flow rate and lower delta T. The estimation of the DEP force magnitude (via numerical simulation of electric field) in comparison to the buoyancy force revealed important understanding of the expected enhancement in the liquid film flow boiling curve when the results were applied to a basic heat transport model. They highlighted that the DEP force was only applicable for a certain range of heat flux when nucleate boiling began, although the DEP electrode was responsible for liquid circulation via EHD conduction mechanism in the low heat flux regime.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Articles in Peer-reviewed Journals	Pearson MR, Yagoobi J. "Experimental study of linear and radial two-phase heat transport devices driven by electrohydrodynamic conduction pumping. " ASME Journal of Heat Transfer. 2015 Feb 1;137(2):022901. http://dx.doi.org/10.1115/1.4025430 , Feb-2015
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Articles in Peer-reviewed Journals	Patel VK, Seyed-Yagoobi J. "Long-term performance evaluation of micro-scale two-phase heat transport device driven by EHD conduction." IEEE Transactions on Industry Applications. 2014 Sep-Oct;50(5):3011-6. http://dx.doi.org/10.1109/IAS.2013.6682448 , Sep-2014
Dissertations and Theses	Patel V. "Terrestrial and Micro-Gravity Studies in Electrohydrodynamic Conduction Driven Heat Transport Systems." Worcester Polytechnic Institute, Worcester. Graduation Date: May 2015. , May-2015
Papers from Meeting Proceedings	Patel VK, Seyed-Yagoobi J. "Recent experimental advances in electrohydrodynamic conduction pumping research." 2014 IEEE Industry Applications Society Annual Meeting, Vancouver, BC, October 5-9, 2014. 2014 IEEE Industry Applications Society Annual Meeting, Vancouver, BC, October 5-9, 2014. http://dx.doi.org/10.1109/IAS.2014.6978363 , Oct-2014
Papers from Meeting Proceedings	Patel VK, Robinson F, Seyed-Yagoobi J, Didion JR. "Liquid Film Boiling Heat Transfer in the Presence and Absence of Gravity." ASME 2014 International Mechanical Engineering Congress and Exposition, Montreal, Quebec, Canada, November 14–20, 2014. In: Proceedings, ASME 2014 International Mechanical Engineering Congress and Exposition, Montreal, Quebec, Canada, November 14–20, 2014. Volume 8B: Heat Transfer and Thermal Engineering: Paper No. IMECE2014-40352, p. V08BT10A074; 7 pages. http://proceedings.asmedigitalcollection.asme.org/proceeding.aspx?articleid=2205402 , Nov-2014

Patents	US8568106 B2. Patent October 2013. https://www.google.si/patents/US8568106 Oct-2013 Seyed-Yagoobi J, Pearson M. "Two-phase heat transport device using electrohydrodynamic conduction pumping."
Patents	Provisional. Provisional patent issued in September 2015. Sep-2015 Seyed-Yagoobi J. "EHD Driven Vapor Compression System."